

Dylan Suzuki, Isaiah Pajarillo, Aidan Hodges, Jackson Muller

CMSI 4072 - Visual Guided Robotic Arm

4/22/2026

Status Report

Accomplishments:

- Implemented inverse kinematics solver using the law of cosines, enabling the arm to compute joint angles for any target coordinate within its reach
- 3D printed a mounting stand for the camera, providing a stable and consistent overhead view
- Developed a bounding box center-point extraction method that converts the camera's detection output into a single (x,y) target coordinate for the IK solver
- Pre-calculated coordinate boundary corrections to compensate for the camera's fisheye lens distortion
- Implemented a keyboard-driven position selection interface that allows the user to choose a target location, which the arm automatically calculates the required joint angles and reaches down to that point
- Implemented a hard-coded grab action that triggers after the arm reaches the target position, allowing the arm to reach down and grasp the object.

Scheduled Tasks:

- Improve coordinate mapping accuracy
- Mitigate fisheye lens distortion

Noteworthy risks, concerns, or problems:

- Fisheye distortion compensation is approximated. The current correction relies on pre-calculated bounds rather than a full map. Objects near the edges of the camera's view have their centers mapped to slightly incorrect coordinates, causing the arm to miss. To mitigate this, we restrict object placement to the central region of the table where distortion error is minimal.
- Bounding box center may not correspond to the object's true graspable point. For irregularly shaped objects, the center may not align with the best grip location, causing failed grabs. To mitigate this we will use predetermined objects.